



Parktown Boys' High School

District: 9

Geography Department

Grade 12 Control Test

Examiner: Mr A Meintjes

Time: 60 Minutes

Moderator: Ms A Clayton

Total: 60 Marks

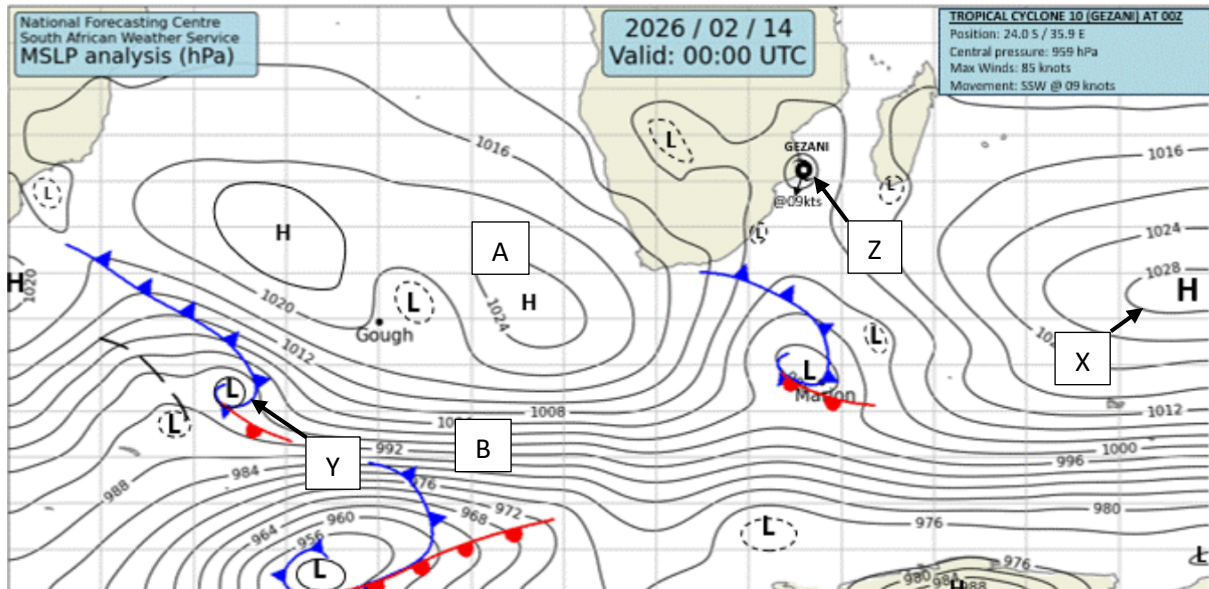
Date: 18 February

INSTRUCTIONS AND INFORMATION

1. The question paper consists of SIX pages.
2. The question paper consists of FOUR questions.
3. ALL diagrams are included in the question paper.
4. Where possible, illustrate your answers with labelled diagrams.
5. Leave a line between subsections answered.
6. Number your answers correctly according to the numbering system used in this question paper.
7. Write neatly and legibly.

Question 1 – Short Questions [15]

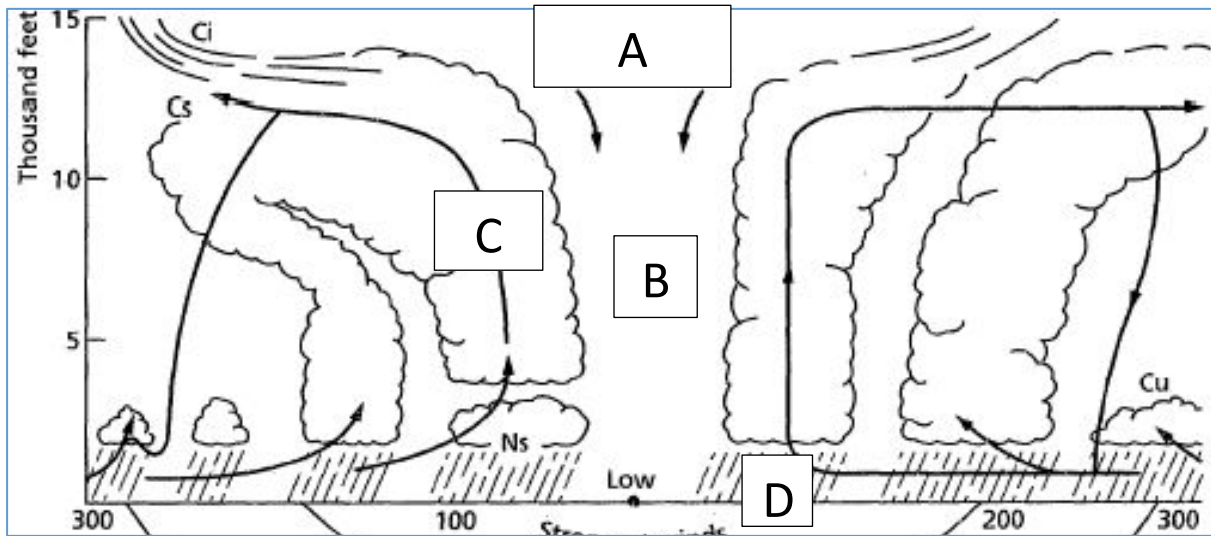
1.1. Refer to the synoptic chart below. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Updog



- 1.1.1. The season shown in the synoptic chart is (Summer/Winter/Autumn/Spring).
- 1.1.2. The high pressure at X is the (South Atlantic / South Indian / Kalahari) high pressure.
- 1.1.3. (Z/Y) is a mid-latitude cyclone.
- 1.1.4. (Z/Y) is a tropical cyclone.
- 1.1.5. The wind speed at B is fast. We know this because (it has a steep pressure gradient / Coriolis force is powerful here / as pressure increase so does the wind speed).
- 1.1.6. The conditions at A are likely (clear and calm / wet and windy / cold and humid).
- 1.1.7. Tropical storm Gezani is the (2nd / 5th / 7th) cyclone of the season.
- 1.1.8. Tropical storm Gezani, according to the chart, is currently over (Mozambique / Madagascar / South Africa).

(8x1) (8)

1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.



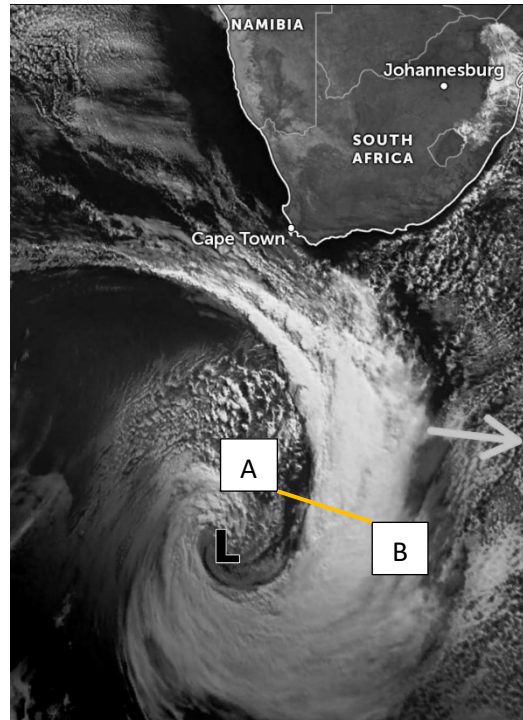
- 1.2.1. The air at A is (diverging / subsiding / low pressure) air.
- 1.2.2. When B is fully formed and visible on satellite imagery, we would say the cyclone is in its (formative / mature / decaying) stage.
- 1.2.3. The weather experienced at B is (violent with heavy rains / clear and calm / extremely windy, but with little to no rain).
- 1.2.4. The clouds that typically form at C are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. These systems occur between latitudes of (0° - 15° S / 5° - 30° S / 30° - 60° S) in the southern hemisphere.
- 1.2.6. The storm surge at D is caused by (tsunamis / very low pressure / heavy rain).
- 1.2.7. Cape Town (always / occasionally / never) experiences hurricane strength winds from these storms.

(7x1) (7)

[Total 15 Marks]

Question 2 – Mid-latitude Cyclones [15]

Refer to the satellite image below and answer the questions that follow.



- 2.1. What phenomenon is being shown in the image? (1x1) (1)
- 2.2. What are the names of the two wind masses that meet at the Polar Front to form mid-latitude cyclones? (2x1) (2)
- 2.3. Why are mid-latitude cyclones more common in winter along the southern coast of South Africa? (1x2) (2)
- 2.4. Draw a fully labelled cross section from A to B. Make sure you have the following:
Front shape
Temperature of air pockets
Cloud types at front (4x1) (4)
- 2.5. Describe how the temperature and pressure in Cape Town is going to change as the mid-latitude cyclone passes over. (3x2) (6)

[Total 15 Marks]

Question 3 – Tropical Cyclones [15]

Refer to the write-up of Tropical Cyclone Gezani below and answer the questions that follow.

The SAWS Monitoring Moderate Tropical Storm Gezani in the South-West Indian Ocean

The South African Weather Service (SAWS) is closely monitoring a moderate tropical storm named Gezani, currently located over the south-west Indian Ocean to the east of Madagascar. At present, there is no indication of any direct threat to South Africa.

Gezani is currently moving westward under favourable environmental conditions, and forecast guidance indicates that landfall over eastern Madagascar is likely on Tuesday afternoon, 10 February 2026. As the system approaches Madagascar, environmental conditions may support further intensification, and Gezani could strengthen into a tropical cyclone, or possibly an intense tropical cyclone, before landfall.

Following landfall, the system is expected to weaken while tracking south-westward across Madagascar. Gezani is then forecast to enter the Mozambique Channel late on Wednesday, 11 February 2026, where environmental conditions may once again become favourable for re-intensification. Some forecast scenarios suggest a possible approach toward the Mozambique coast later in the week; however, uncertainty remains regarding the exact track and intensity during this phase.

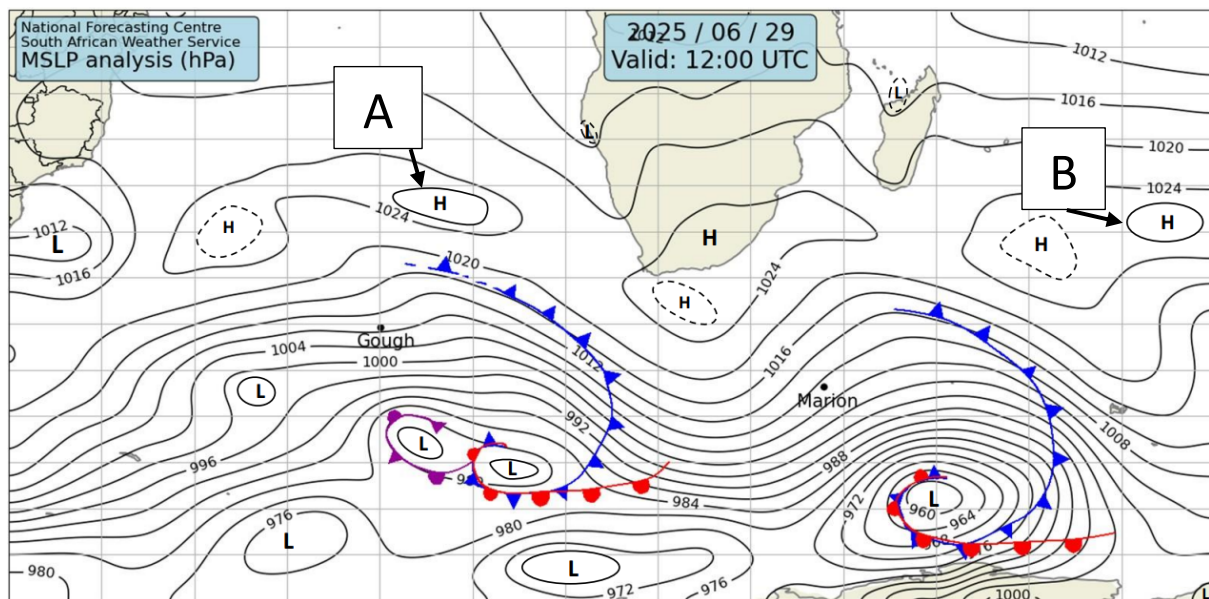
The SAWS advises vessels operating in or transiting through the south-west Indian Ocean and the Mozambique Channel, as well as offshore marine operations to exercise caution, as rough seas, strong winds, and hazardous marine conditions are expected in the vicinity of the storm.

- 3.1. Which ocean did Cyclone Gezani pass through while moving towards Mozambique? (1x1) (1)
- 3.2. According to the article, why should vessels operating in the south-west Indian Ocean and Mozambique Channel exercise caution? (2x1) (2)
- 3.3. Gezani is expected to weaken as it passes over Madagascar. Provide TWO reasons as to why this will happen. (2x2) (4)
- 3.4. Discuss TWO economic impacts that Gezani would have on Madagascar, and provide TWO solutions that Madagascar could implement to reduce the impact of future cyclones. (4x2) (8)

[Total 15 Marks]

Question 4 – Anticyclones [15]

Refer to the synoptic chart below and answer the questions that follow.



- 4.1. What season is depicted in the synoptic chart? Provide TWO pieces of evidence to substantiate your answer. (1+2) (3)
- 4.2. List TWO characteristics of an anticyclone. (2x1) (2)
- 4.3. Name the anticyclone present over South Africa. (1x1) (1)
- 4.4. Name the TWO anticyclones visible at A and B on the chart. (2x1) (2)
- 4.5. What is the actual pressure in hectopascals (or millibar) in at A? (1x1) (1)
- 4.6. Explain why all of these anticyclones are situated here at this time. (1x2) (2)
- 4.7. Explain how the position of the anticyclones over South Africa and mid-latitude cyclones depicted in the chart lead to typical weather conditions across South Africa during this season. (2x2) (4)

[Total 15 Marks]

GRAND TOTAL: 60 MARKS